

REAL-TIME SCORE FOLLOWING IN SHEET MUSIC IMAGES AS SEQUENTIAL CANDIDATE SELECTION

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ABSTRACT

Score following aligns a live performance to its score in real time, committing to a position at every instant with no lookahead. The performance is audio and the score is an image, and most systems resolve that mismatch by collapsing the two into a single modality before aligning: optical music recognition converts the page into symbols, or transcription converts the audio into notes, and the alignment is then carried out within whichever modality survives. Each detour inherits the errors of its conversion, and optical music recognition in particular remains unreliable on real scores. Sheet-image score following keeps both modalities and learns the correspondence directly, and its strongest systems condition a visual detector on the audio. What limits them is annotation: training needs performances aligned to notehead coordinates on the page, costly enough that nearly all of it is synthesized, so these systems work on synthesized audio and lose much of their accuracy on real recordings. Closing that gap is the field’s central open problem, and every remedy proposed for it has treated the gap as a failure of perception. We show it is not. On real recordings a conditional YOLO detector already proposes a correct position at every scored onset, and an oracle under the same real-time constraints, committing at each instant and never moving backwards, reaches 99.6% where the deployed rule reaches 80.0%. We therefore recast the task as sequential selection over the candidates the detector already emits, combining a parameter-free transition model with a set-based scorer that judges each candidate against its competitors in the same instant. On real piano recordings this tracks 94.4% of onsets within 0.5 s against 88.3% for the best published system, and holds that margin across three independently trained detectors. Analysis of the remaining error shows that the image representation is irrelevant while the temporal history is decisive, that accuracy on synthesized audio barely predicts accuracy on real audio, and that the forward-motion assumption fails at repeats.

Index Terms— Score following, sheet music, audio-to-score alignment, real-time, music information retrieval

1. INTRODUCTION

A musician plays from a printed page; a machine listens, looks at the same page, and at every moment reports where in the music the performer has got to. That is *score following* [1], and it is what automatic page turning [2], live computer accompaniment [3, 4, 5] and score-synchronised visualisation are built on.

Real time is the constraint. Aligning a performance to its score offline is a different problem, usually called audio-to-score alignment or music synchronisation: the whole recording is in hand, a global method such as dynamic time warping settles every position at once [6], and even repeats and jumps can be resolved after the fact [7, 8]. Offline alignment underpins cross-modal retrieval and piece identification, where a recording is matched against a collection of scores [9]. Score following, also called music tracking [10], gets none of that: it sees only what has been played, commits to a position immediately, and never revises it. Everything below respects this constraint, including the oracles we use to measure what is achievable.

Two modalities. The remaining difficulty is that the two sides are different kinds of object. The performance is audio, the score is an image, and there is no shared space in which to compare them. The literature answers this in three ways.

1.1. Related work

Convert the score. Optical music recognition turns the page into symbols, and a classical follower tracks the performance over a discrete set of score states. This line is mature: online dynamic time warping [6], duration-focused probabilistic models [11], hidden semi-Markov models absorbing performer errors and arbitrary repeats [12], particle filters for rests and tempo change [13], switching state-space models with hidden tempo [14], and multi-agent tracking for robustness to arbitrary deviation [10]. Its weakness is the conversion: optical music recognition remains error-prone on real scores [15] and the follower inherits every mistake.

Convert the audio. The symmetric route transcribes the performance into notes and compares symbol to symbol. Recent work pairs real-time piano transcription with symbol-level tracking and reports strong accuracy [16], but the follower is

then bounded by the transcriber, which restricts it to instruments transcription handles well.

Convert neither. Sheet-image score following learns the correspondence between audio and pixels directly. Dorfer et al. [17] first classified position into staff-level buckets on image crops; reinforcement learning then framed tracking as an agent navigating a score strip [18, 19]; CUNet [20] moved to full pages, emitting an audio-conditioned segmentation whose centre of mass gives the position; and CYOLO [21] recast it as audio-conditioned object detection, with CYOLO-SB [22] adding heads for the enclosing system and bar. The concurrent CODA [23] cascades system, bar and note selection over the layout regions of an annotated score, with beam search and learned transition penalties. Beneath these differences the field has settled on one visual front end: a convolutional network over the page whose feature maps are modulated by feature-wise linear modulation [24], $f_{\text{FiLM}}(x) = s(z) \odot x + t(z)$, from a vector z encoding the recent audio. Our experiments use three separately trained instances of it, so the conclusions are about the family and not one checkpoint.

The gap. Training any of these needs performances aligned to notehead coordinates on the page, an annotation costly enough that nearly all of it is synthesized: a score is rendered to MIDI, the MIDI to audio, and the alignment comes for free. The result is systems that work on synthesized audio and lose much of their accuracy on a real instrument in a real room. CUNet [20] falls from 85.5% of onsets tracked within 0.5 s to 22.4%, CYOLO-SB [22] from 89.3% to 79.9%, and CODA [23] from 95.5% to 88.3%. Closing this gap is the field’s central open problem, and every remedy proposed for it has been a perception remedy: reverberant augmentation [21], extra proprietary recordings [22], or a new architecture [23].

1.2. Perception is not the bottleneck

We tested that premise before accepting it. A conditional detector does not produce a position directly; it scores thousands of candidate noteheads and reports the most confident one. Keeping the 256 most confident, we find that on real room-microphone recordings a candidate within the 0.5 s tolerance is present at *every one* of 4,149 scored onsets. An oracle free to pick any of them is therefore perfect, but could not be run live; the one that matters must commit at each instant and never move backwards, exactly as a deployed follower must, and it still reaches 99.6%. The deployed rule reaches 80.0% (Fig. 1). Nineteen of the twenty points lost to real audio survive the real-time constraint and are recoverable from what the model already saw. The room degrades the confidences, not the evidence.

We therefore recast sheet-image score following as a sequential decision over the positions the page affords, rather than the regression of a coordinate. CANDOR makes that decision with two parts: a transition model scoring a displacement against how much time has passed, and a set-based scorer judg-

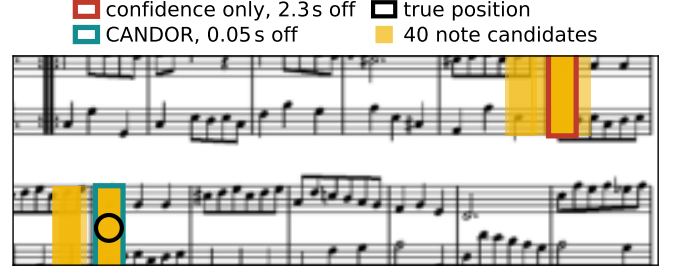


Fig. 1. One instant of a real room-microphone recording. Yellow: 40 of the candidate positions the detector scores. The most confident (red) lies on the wrong staff, 2.3 s from the truth; CANDOR (teal) takes a less confident candidate on the true note (ring). The correct position was available and was not chosen, which is the characteristic failure on real audio.

ing each candidate against its competitors in the same instant, trained listwise on synthesized audio alone. The decision is causal and greedy, with no lookahead and no beam, and costs 1.25 ms per instant on a single CPU thread against the 50 ms available at 20 frames per second.

Our contributions are: (i) evidence, from candidate-set oracles, that this gap is a decision failure and not a perception failure; (ii) CANDOR, which acts on it and sets a new best on real recordings by a margin holding across three independently trained perception models; (iii) an analysis of what remains, covering the irrelevance of the image representation, the unreliability of synthesized audio for model selection, and the failure at discontinuities.

2. CANDOR

2.1. Candidate positions

A performance is processed in instants $t = 1, \dots, T$. At each instant the perception model reads the audio so far and the current page and scores note boxes $b_i = (x_i, y_i, w_i, h_i)$ with confidence o_i ; we keep the $K = 256$ most confident, C_t . Staves are concatenated along one horizontal axis, the unrolled coordinate u_i , so positions across a line break are adjacent. The released detectors score 8,112 boxes per instant and report

$$\hat{i}_t = \arg \max_{i \in C_t} o_i. \quad (1)$$

CANDOR replaces Eq. 1 and reads only C_t and its own previous choice $\hat{i}_{t-1}$.

2.2. Transition model

Successive decisions must describe a plausible trajectory. Between instants Δ_t frames apart the performer advances by a roughly predictable amount, so the displacement $d_i = u_i - \hat{u}_{t-1}$ is scored by a floored Gaussian whose mean grows with

elapsed time,

$$\log \pi_i = \max \left(-\frac{(d_i - \mu \rho_t)^2}{2\sigma^2}, J \right), \quad (2)$$

$\rho_t = \text{clip}(\Delta_t/\Delta_0, 0.2, 8)$ with $\Delta_0 = 5$ frames, the mean inter-onset gap in training. The floor J keeps a jump possible at bounded cost. Combined with confidence, $h_i = \log o_i + \log \pi_i$ already recovers a large part of the gap on its own.

2.3. Set-based scorer

The transition model treats candidates independently given the history. What it cannot express is comparative: whether *this* box, among those on offer in this instant, looks like a real notehead rather than a spurious response. Each candidate is described by $\phi_i \in \mathbb{R}^{33}$, holding its confidence relative to the instant’s best, box geometry, motion relative to the history, relation to any coarse regions the model emits, and a neighbourhood group (how many candidates fall within 25 and 100 px, spacing to the nearest on either side, rank in reading order, size against the instant’s median). We also read f_i , the 128-dimensional FiLM-conditioned feature from which the detection head scored the box, so the decision sees the representation perception used. A permutation-invariant network [25], as in learned re-scoring of detections [26], embeds the set and pools a context shared by all candidates:

$$e_i = g([\phi_i; \text{ReLU}(W f_i)]), \quad c = \left[\frac{1}{K} \sum_j e_j; \max_j e_j \right], \quad (3)$$

$$s_i = r([e_i; c]), \quad (4)$$

with g, r two-layer perceptrons (64, 32 and 64, 1 units) and $W \in \mathbb{R}^{64 \times 128}$. The maximum is taken per dimension, so every candidate is judged against its competitors rather than in isolation.

2.4. Training and inference

The decision is supervised listwise. With ε_i the error in frames of candidate i , the target is $q_i \propto \exp(-\varepsilon_i/\tau)$, $\tau = 3$, and

$$\mathcal{L} = - \sum_{i \in \mathcal{C}_t} q_i \log \frac{\exp s_i}{\sum_j \exp s_j}, \quad (5)$$

so the scorer learns to order a set rather than to regress a coordinate. History features come from previous decisions, perturbed on 30% of samples by Laplace noise of scale 30 px to expose training to the drift of sequence prediction [27]. At inference CANDOR decides

$$\hat{t}_t = \arg \max_{i \in \mathcal{C}_t} \beta s_i + (1 - \beta) h_i, \quad (6)$$

causally and greedily, one pass over 256 candidates per instant, in 1.25 ms on a single CPU thread against the 50 ms available at 20 fps. Training on the residual $s_i - h_i$ and deciding $\arg \max_i (s_i + h_i)$ is Eq. 6 at $\beta = 0.5$ and eliminates all four constants.

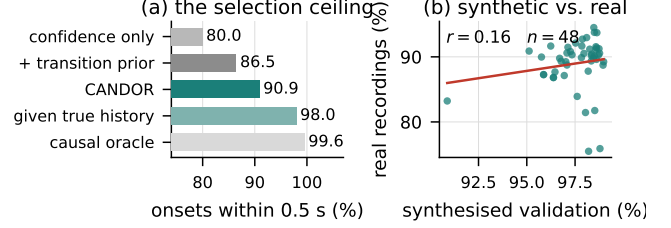


Fig. 2. (a) Every step from the deployed rule to the ceiling on the real recordings; CANDOR to *given true history* is drift, above it is ranking. (b) Each point is one of 48 trained models: accuracy on synthesised validation audio against accuracy on the real recordings.

3. EXPERIMENTAL SETUP

Data. The decision model is trained on 353 pieces of MSMD [9], synthesized and convolved with measured room impulse responses [28], and validated on 19, using AdamW ($3 \cdot 10^{-3}$, cosine, 40 epochs, batch 256). The evaluation set is the real-performance benchmark of [19]: 16 pieces, 25 pages and 4,149 onsets on a Yamaha AvantGrand N2 captured with a room microphone. A held-out set of 80 further pieces is rendered with a real impulse response and noise from 12 to 0.5 dB SNR.

Metric. The ratio of onsets whose predicted position maps to within 0.05 to 5 s of the true time, 0.5 s primary, with 95% intervals bootstrapped over pieces [29].

Constants. (μ, σ, J, β) are fitted leave-one-piece-out: for each evaluation piece they are chosen on the other 15, pieces weighted equally, so none influences its own decoding. The residual variant needs no fit at all. Seeds move results by about a point, so rows are seed means with deviations.

4. RESULTS

4.1. A new best on real recordings

Table 1 sets CANDOR against the published literature on the same recordings. It tracks 94.4% of onsets within 0.5 s, where the previous best was 88.3%, and leads at every threshold. The margin is not an artifact of one perception model: retrained on whichever candidates each model emits, the decision gains 20.4, 10.7 and 7.9 points on three of them, and on the weakest, whose training data is entirely public, it passes the point at which the strongest one begins. A decision model does not need good proposals, only proposals containing the answer. On the 94 synthetic test pieces CANDOR reaches .955 where argmax reaches .893, and on 80 held-out pieces under a real impulse response and noise it beats the transition model at every level from 12 to 0.5 dB.

The tightest thresholds qualify this: on CYOLO-SB candidates CANDOR reaches .690 and .714 at 0.05 and 0.1 s, below

Table 1. Real piano recordings: 16 pieces, 4,149 onsets, Yamaha AvantGrand N2 captured with a room microphone. Ratio of onsets tracked within each error threshold. [†] perception model trained with additional proprietary recordings, public checkpoint. *Layout*: system and bar annotations required at test time. Δ is the gain at 0.5 s over the perception model the row is built on. Published figures as reported in [23].

Method	Onset error threshold (s)					Layout	Δ
	≤ 0.05	≤ 0.1	≤ 0.5	≤ 1.0	≤ 5.0		
MM-Loc [17]	.364	.406	.585	.611	.735	no	
RL agent [19]	.185	.200	.476	.603	.901	no	
CUNet [20]	.113	.125	.224	.266	.443	no	
CYOLO [21]	.563	.581	.712	.749	.919	no	
CYOLO-SB [22]	.610	.630	.799	.832	.960	no	
CYOLO-SB+A [†] [22]	.682	.706	.865	.891	.981	no	
CODA [23]	.721	.743	.883	.914	.987	yes	
CANDOR on CYOLO	.726	.747	.916	.950	.991	no	+20.4
CANDOR on CYOLO-SB	.690	.714	.906	.942	.982	no	+10.7
CANDOR on CYOLO-SB+A [†]	.760	.785	.944	.969	.996	no	+7.9

Table 2. Components of CANDOR, cross-validated per piece on the real recordings at 0.5 s. *argmax* is Eq. 1 over the same candidates, *+transition* adds Eq. 2, *full* adds the scorer.

	<i>n</i>	<i>argmax</i>	<i>+transition</i>	<i>full</i>
<i>Perception model</i>				
CYOLO [21]	3	71.3	72.5	91.6 $\pm$ 1.4
CYOLO-SB [22]	6	80.0	86.5	90.6 $\pm$ 0.9
CYOLO-SB+A [†]	3	86.5	90.8	94.4 $\pm$ 0.9
<i>Decision model, on CYOLO-SB candidates</i>				
residual, no fitted constants	3	80.0	86.5	88.7 $\pm$ 0.9
without temporal history	3	80.0	86.5	83.7 $\pm$ 0.4
<i>Image features f_i read by the scorer</i>				
FiLM-conditioned backbone	6	80.0	86.5	90.6 $\pm$ 0.9
DINOv2 [30]	1	80.0	86.5	90.9
none	1	80.0	86.5	90.9

CODA’s .721 and .743, because a decision is only as precise as the boxes it chooses between. Deciding recovers which notehead, not where within it.

Table 2 separates the parts: the transition model alone, with no learned parameters, accounts for roughly half the gain, and the residual variant, which fits no constant to any evaluation data, still reaches 88.7%.

4.2. What the decision depends on

Not the image. Giving the scorer the FiLM-conditioned backbone feature, DINOv2 patch features [30] on the same grid, or no image feature at all yields 90.6, 90.9 and 90.9, a spread well inside the 0.9-point seed deviation. On synthesized validation audio the same three differ by 3.4 points, so those features are learnable and useful there and worth nothing on real audio. **Nor the encoders.** We rebuilt the perception model with a different audio encoder twice: a frozen MERT-95M tower [31] with only a projection trained, which reaches 46.2 against 90.6 and overfits 12.6-fold on the 353 training pieces, and the two-

Table 3. Recovery from 41 discontinuities spliced into the real recordings, on CYOLO-SB+A candidates. *rec@1*: fraction first tracked correctly within 1 s. *post*: onsets tracked within 1 s over the following 5 s.

Decision rule	break present		no break	
	rec@1	post	rec@1	post
argmax (1)	0.27	0.27	0.15	0.26
CANDOR	0.00	0.08	0.00	0.09
+ silence trigger	0.15	0.17	0.00	0.09
+ confidence trigger	0.10	0.26	0.02	0.23

layer causal Mamba of [23] trained from scratch as they train it, which reaches .. The two are not matched on pretraining and fail differently, on generalisation and on capacity, arriving at the same place. Replacing the *visual* stem with frozen DINOv2-base patches on the same 52×52 grid, against a control retrained identically, gives ..

The history. Removing the features computed against the previous decision costs 7.2 points, the largest single ablation here. Teacher forcing attributes two thirds of the remaining error to drift: given the true previous position the same scorer reaches 98.0% against the causal oracle’s 99.6% (Fig. 2a). The sequence is hard, not the image.

Synthesized audio does not predict real audio. Across 48 trained models, accuracy on the synthesized validation split correlates $r = 0.16$ with accuracy on the real recordings (Fig. 2b); the best-validation model scores 89.3 on real audio, below one reading no image features at all. A reverberant held-out set does better ($r = 0.80$) and still mispicks. Since the whole field selects models on synthesized audio, this is a caution beyond our own experiments, and it is why our constants are cross-validated on the target recordings instead.

4.3. Where the forward assumption fails

Eq. 2 assumes the performance moves forward, which is why the decision is tractable and also why a repeat or a jump breaks it. Following the protocol of [23] we splice three discontinuities into each performance at random destinations, re-running perception on the spliced audio so its conditioning is genuinely wrong afterwards, and excluding jumps across a page boundary where state resets anyway, leaving 41.

Table 3 reports a systematic failure. After a jump CANDOR is far worse than the rule it replaces, 0.08 against 0.27 of onsets tracked over the next five seconds, because the transition model holds the trajectory at a stale position. Suppressing it on detected silence helps where a break exists and not otherwise, as its design implies; triggering on the scorer’s own confidence needs no break and recovers most of the loss, but never beats argmax; the same ordering holds over 220 jumps on the synthetic benchmark. A transition model strong enough to be worth ten points on continuous playing is strong enough to be wrong at a discontinuity, and reconciling the two is open.

5. CONCLUSION

The synthetic-to-real gap in sheet-image score following has been read as a perception failure. It is a decision failure: a correct position is already among the candidates at every onset, and choosing among them in real time takes real piano recordings from 80.0% to 94.4%, against 88.3% previously.

Limitations. The correct page is given, all performances are piano, and 16 evaluation pieces cannot resolve differences of a point.

6. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical study that uses only publicly available data; no ethical approval was required.

7. ACKNOWLEDGMENTS

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